

Smart Homes

Comparison of ZigBee, Z-Wave, and Thread for Home Automation

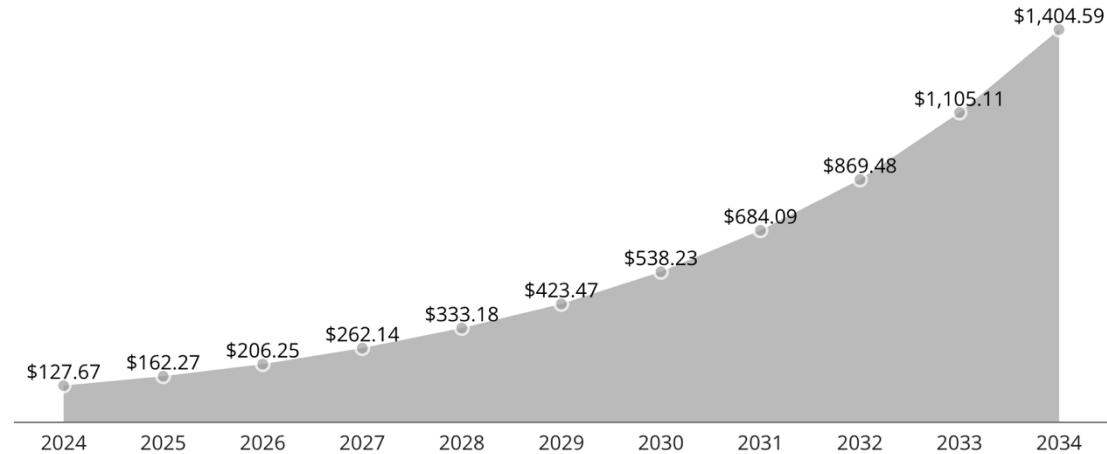
Matt Krueger | Topics in Computer Science II: Real-Time Cyber-Physical Systems | Fall 2025

Background

Growing industry with lots of benefits!

Smart Home Market Size

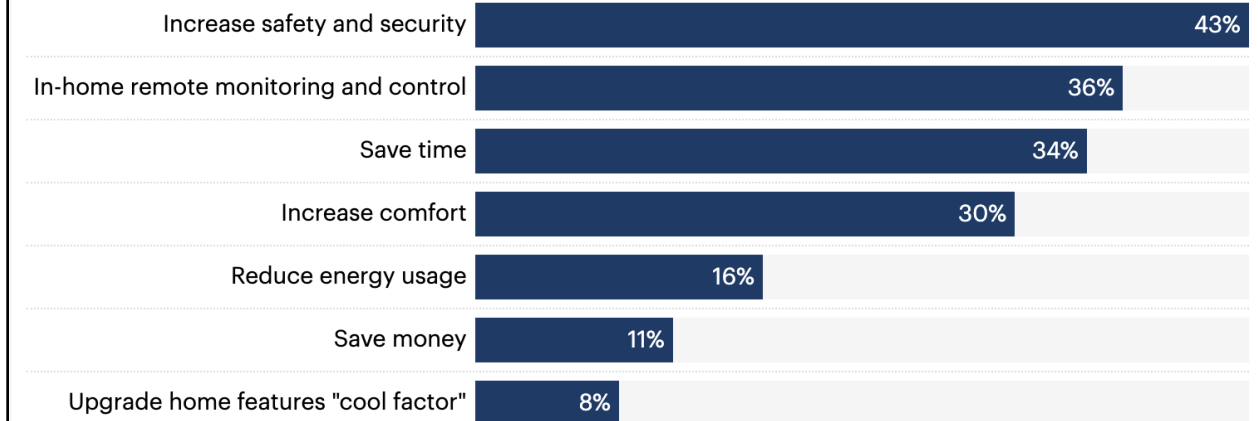
USD Billion



Source: Precedence Research

Today's
Homeowner
with Danny Lipford
25 YEARS

Most common benefits of smart devices, 2023



Source: Deloitte Insights

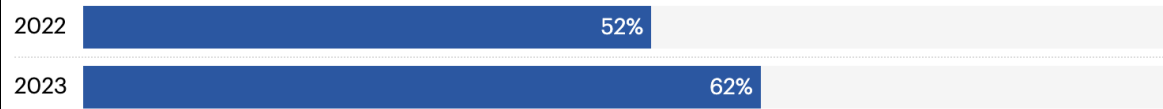
 **ConsumerAffairs**

Installing a smart home system costs between \$205 and \$1,659, according to **Angi**.

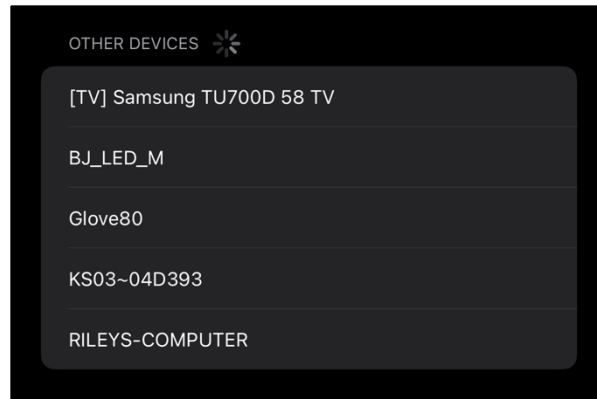
Background

Digital == lots of attack vectors!

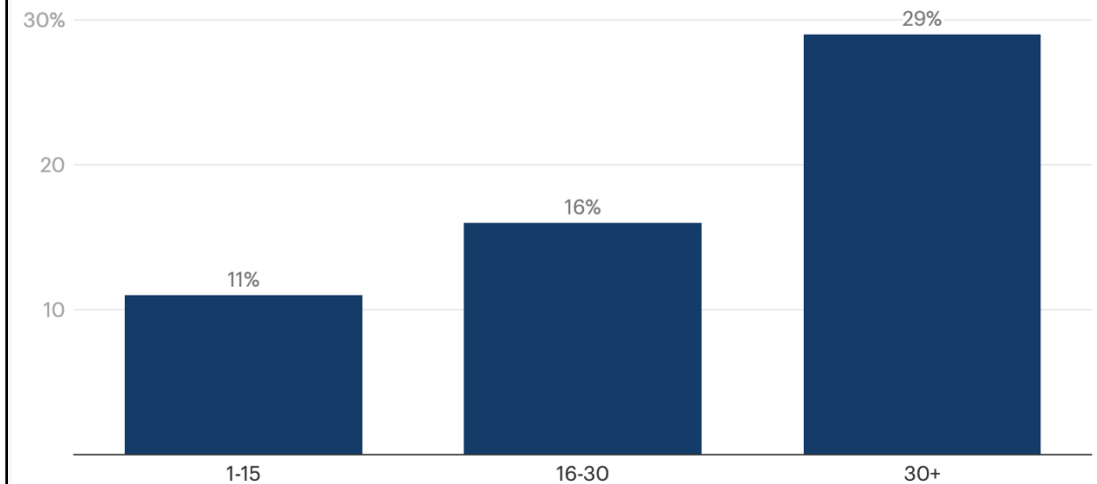
Smart home users concerned about security and privacy of their devices' data, 2022-2023



Source: Deloitte Insights



Households who faced at least two data breaches over the previous 12 months, by number of devices, 2023



Source: Deloitte Insights



Motivation

Personal Interest



Project

Comparison of ZigBee, Z-Wave, and Thread for Home Automation



vs



vs



Goals

Learn About Industry Leaders

- Architecture
- Applications
- Features

Compare Protocols

- Performance
- Security
- Interoperability

Plan

1. Gather Information
2. High-Level Comparison
3. Present Results

Plan

1. Gather Information

Conferences > 2017 IEEE 13th International ...

Analysis on the Security Architecture of ZigBee Based on IEEE 802.15.4

Publisher: IEEE [Cite This](#) [PDF](#)

Bo Fan [All Authors](#)

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Conferences > 2015 International Conference...

Performance analysis of ZigBee based real time Home Automation system

Publisher: IEEE [Cite This](#) [PDF](#)

Jaypal Baviskar ; Afshan Mulla ; Manish Upadhye ; Jeet Desai ; Aniket Bhovad [All Authors](#)

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Abstract

Document Sections

I. Introduction

II. Factors Influencing

Performance Analysis of the
Implemented System

III. Implemented System
Methodology

IV. Performance Analysis of
the Zigbee Based Has

V. Conclusion

Authors

Abstract:

Incorporation of wireless technology in Automation system has led to the development of remote monitoring and controlling of loads. It is essential to select and implement an appropriate technology according to the desired application area. This paper describes the various factors viz. network topology, network span and transmission time, which influence the performance of the system. It also illustrates ZigBee (over IEEE 802.15.4) based Home Automation system (HAS), for remote controlling and monitoring of loads/appliances. In order to estimate the total power consumption of the desired load, the designed HAS can be operated in either of the two modes viz. Energy measurement IC 'CS5490' based mode and Current sensor IC 'WCS 2705' based mode. It provides real time power monitoring with the help of a PC-based GUI application developed on Net Beans Java platform. Further, the paper emphasizes on performance evaluation of the system by analyzing the various performance metrics viz. the Latency, Round Trip Delay (RTD) time and Received Signal Strength Indicator (RSSI).

Published in: 2015 International Conference on Communication, Information & Computing Technology (ICCICT)

Date of Conference: 15-17 January 2015

DOI: 10.1109/ICCICT.2015.7045685

Date Added to IEEE Xplore: 23 February 2015

Publisher: IEEE

Abstract:

With the rapid development of the network technology and the wireless communication technology, the application of ZigBee is increasing wide application, which is an kind of emerging technology in the field of wireless communication. ZigBee is becoming more and more significant. In this paper, ZigBee security architecture and security components, security keys and the Trust Center, security measures of each layer are displayed the steps of security implement of ZigBee.

Published in: 2017 IEEE 13th International Symposium on Autonomous Decentralized Systems (ISADS)

Date of Conference: 22-24 March 2017

DOI: 10.1109/ISADS.2017.8045685

Date Added to IEEE Xplore: 08 June 2017

Publisher: IEEE

Conference Location: B

SECTION I.


zigbee

Conferences > 2021 IEEE International Confe...

New LDoS Attack in Zigbee Network and its Possible Countermeasures

Publisher: IEEE [Cite This](#) [PDF](#)

Satoshi Okada ; Daisuke Miyamoto ; Yuji Sekiya ; Hiroshi Nakamura [All Authors](#)

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Cites in
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Abstract

Document Sections

I. Introduction

II. Overview of Zigbee

III. Related Work

IV. New Attack Scenarios

V. Proposed Countermeasures

Show Full Outline

Authors

Figures

References

Abstract:

Low-Rate DoS (LDoS) attacks degrade the quality of service with less traffic than ordinary DoS attacks. LDoS attacks can easily evade conventional counter-DoS detection mechanisms because their time-averaged flow is small. Thus, LDoS attack is getting a serious problem nowadays. With the recent spread of IoT devices, Zigbee, one of the IoT communication standards, attracts much attention. Zigbee is a low-power wireless communication protocol at the sacrifice of its transfer range and bandwidth. Since Zigbee consumes low power, it is widely adopted for inexpensive small sensors and IoT that run on batteries. The advantage of the low power consumption of Zigbee is due to its use of indirect transmission. However, we newly found that vulnerability to LDoS attacks existed in the indirect transmission. In this study, we first point out this vulnerability and describe new LDoS attack scenarios exploiting it. Next, we propose simple countermeasures against them which require low computational cost. Then, simulation experiments are conducted to evaluate the impact of the attacks and the effectiveness of the proposed countermeasures. The experimental results show that the new LDoS attacks can decrease the normal packet arrival rate to 0% by adjusting the timing of attacks. It is also revealed that our proposed countermeasures successfully prevent such cases and improve the minimum arrival rate to 80% at best.

Published in: 2021 IEEE International Conference on Smart Computing (SMARTCOMP)

1. Gather Information

Conferences > 2016 International Conference...
Smart homes automation using Z-wave protocol
 Publisher: IEEE Cite This PDF
 Muneer Bani Yassein ; Wail Mardini ; Ashwaq Khalil All Authors
 68 Cites in Papers 3710 Full Text Views
 Abstract Document Sections
 I. Introduction
 II. Z-Wave Technology
 III. Z-Wave Door Locks Scenario and Security
 IV. Z-Wave Light Switch Scenario
 V. Conclusion and Future Work
 Authors Figures
 Abstract:
 The rapid development of information technology and computer networks make them part of almost everything in our daily life and it became impossible to abandon their use. One of the main and important applications of technology in homes is home automation including controlling and automation of electronic and electrical machines remotely. Wireless Home Automation Networks (WHANs) are used in homes to connect the different devices together and to the Internet. In order to control home devices remotely, there are many popular protocols such as INSTEON, ZigBee, and Home Plug. In this paper, we focus on the relatively new protocol called Z-Wave protocol and we discuss its development and applications in smart homes. This wireless protocol has many advantages over the popular and widely used ZigBee protocol as it provides better reliability, low radio rebirth, easy usage, and easy Interoperability.
 Published in: 2016 International Conference on Engineering & MIS (ICEMIS)
 Date of Conference: 22-24 September 2016 DOI: 10.1109/ICEMIS.2016.7745306
 Date Added to IEEE Xplore: 17 November 2016 Publisher: IEEE
 ISBN Information: 9781467394494 Conference Location: Agadir, Morocco

ZCover: Uncovering Z-Wave Vulnerabilities Through Systematic Security Analysis of Application Layer Implementation

Publisher: IEEE

Cite This

PDF

Nkuba Kayembe Carlos ; Jimin Kang ; Seunghoon Woo ; Heejo Lee All Authors

41 Full Text Views

Abstract

Document Sections

I. Introduction

II. Background, threat model and challenges

III. Methodology

IV. Evaluation

V. Discussion

Show Full Outline ▾

Authors

Figures

Abstract:

The increasing use of smart home technologies has raised concerns about security vulnerabilities, particularly in Z-Wave systems. Existing approaches hold the promise of assessing Z-Wave security in slave devices but fall short of being effectively applied to discover vulnerabilities in Z-Wave controllers, which are central to Z-Wave systems. We present ZCover, a framework for systematically analyzing the application layer of Z-Wave controllers to uncover security vulnerabilities. By extracting the known and unknown properties of the Z-Wave controller and utilizing mutation that considers the correlations of the Z-Wave packet frame fields, ZCover can effectively discover unknown vulnerabilities in the target Z-Wave controller. Evaluation on nine real-world Z-Wave devices showed that ZCover outperformed existing Z-Wave security research, by discovering 15 previously unknown critical vulnerabilities with 12 new CVE IDs assigned. ZCover can be utilized as a resource for ensuring the security of Z-Wave controllers in building a secure Z-Wave smart home.

Published in: 2025 55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)

Date of Conference: 23-26 June 2025DOI: 10.1109/DSN64029.2025.00068

Date Added to IEEE Xplore: 11 July 2025Publisher: IEEE



Plan

1. Gather Information

Journals & Magazines > IEEE Access > Volume: 7

Latency Analysis of Wireless Networks for Proximity Services in Smart Home and Building Automation: The Case of Thread

Publisher: IEEE Cite This PDF

Dapeng Lan · Zhiho Pang · Carlo Fischione · Yu Liu · Amir Taherkordi · Frank Eliassen All Authors



AN1141: Thread Mesh Network Performance

This application note details methods for testing Thread mesh network performance. With an increasing number of mesh networks available in today's wireless market, it is important for designers to understand the different use cases among these networks and their expected performances. When selecting a network or device, designers need to know the network's performance and behavior characteristics such as battery life, network throughput and latency, and the impact of network size on scalability and reliability.

This application note demonstrates how the Thread mesh network differs in performance and behavior from other mesh networks. Tests were conducted using Silicon Labs' Thread Mesh software stacks and the Wireless Gecko SoC platform capable of running Thread Mesh and Proprietary protocols. The test environment was a commercial office building with active Wi-Fi and Zigbee networks in range. Wireless test clusters were deployed in hallways, meeting rooms, offices, and open areas. The methodology for performing the benchmarking tests is defined so that others may run the same tests. These results are intended to provide guidance on design practices and principles as well as expected field performance results.

Additional performance benchmarking information for other technologies is available at <http://www.silabs.com/mesh-performance>.

The information in this document is based on the SLThread implementation of Thread. SLThread reached 'end of service' in December 2019. Silicon Labs is replacing SLThread with an implementation of the more popular OpenThread. We anticipate that the results from OpenThread will be very close to the SLThread results.

KEY POINTS

- Wireless test network in Silicon Labs Research and Development (R&D) is described.
- Wireless conditions and environments are evaluated.
- Mesh network performance including throughput, latency, and large network scalability is presented.

Abstract:

Proximity service (ProSe), using the geographic location and device information by enriches the services we use to interact with people and things around us. ProSe has proximity and also in smart home and building automation (Google Home). To enable network protocols and communication infrastructures are needed. Thread is a new building automation (BA), which supports mesh networks and native Internet protocols should be carefully studied when used in user-friendly and safety-critical ProSe in smart level model of latency in the Thread mesh network is presented. The accumulated latency from the application layer to the physical layer. A Markov chain model is used to derive medium access control service time. The system level model is experimentally validated. The outcomes show that the system model results match well with the experimental model, a software tool is developed to estimate the latency of the Thread mesh network information to develop user-friendly and safety-critical ProSe in smart home and BA.

THREAD



UG103.11: Thread Fundamentals

This version of UG103.11 has been deprecated.

For the latest version, see docs.silabs.com.

This document includes a brief background on the emergence of Thread, provides a technology overview, and describes some key features of Thread to consider when implementing a Thread solution.

Silicon Labs' *Fundamentals* series covers topics that project managers, application designers, and developers should understand before beginning to work on an embedded networking solution using Silicon Labs chips, networking stacks such as EmberZNet PRO or Silicon Labs Bluetooth, and associated development tools. The documents can be used as a starting place for anyone needing an introduction to developing wireless networking applications, or who is new to the Silicon Labs development environment.

KEY POINTS

- Introduces Thread and provides a technology overview.
- Describes some of the key elements of Thread, including its IP stack, network topology, routing and network connectivity, joining a network, management, persistent data, security, border router, device commissioning, and application layer.
- Contains updates for Thread Specification 1.3.0.
- Includes next steps for working with the Silicon Labs OpenThread offering.

Plan

2. High-Level Comparison

- Decision Matrix

Metrics:

- Smart Home Automation?
- Building Automation?
- Real-Time?
- Provides SDK?
- Vendor Specific?
- Device Versatility?
- Cost?
- Security?
- Market Share?

metrics are subject to change

Plan

3. Present Results



References

Information

- [Today's Homeowner](#)
- [IEEE Explore](#)
- [Silicon Labs](#)
- [Consumer Affairs](#)

Images

- [ZigBee Logo](#)
- [Thread Logo](#)
- [Z-Wave Logo](#)
- [Weight Module](#)
- [ESP32](#)
- [Arduino](#)
- [ESP8266](#)
- [Raspberry Pi 4](#)
- [LED Strip](#)
- [Question](#)